

THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

Screening Test - Kaprekar Contest

(NMTC- at **SUB-JUNIOR LEVEL** VII to VIII Standards)

Saturday, 28th August 2010.

Note:

- 1) Fill in the response sheet with your Name, Class, the institution through which you appear in the specified places.
- 2) Diagrams are only visual aids; they are not drawn to scale.
- 3) You are free to do rough work on separate sheets.
- 4) Duration of the test: 2 p.m. to 4. p.m.- 2 hours.

PART – A

Note:

- Only one of the choices A,B,C,D is correct for each question. Shade that alphabet of your choice in the response sheet. (If you have any doubt in the method of answering, seek the guidance of your supervisor).
- For each correct response you get 1 mark; **for each incorrect response you lose $\frac{1}{4}$ mark.**

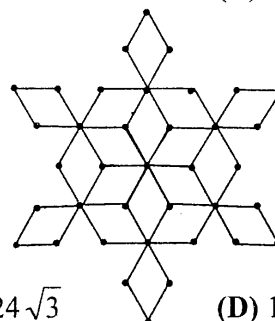
1. Given a sequence of two digit numbers grouped in brackets as follows (10), (11, 20), (12, 21, 30), (13, 23, 31, 40) (89, 98), (99). The digital sum of the numbers in the bracket having maximum numbers is

(A) 9 (B) 10 (C) 9 or 10 (D) 18

2. Using the digits 2 and 7, and addition or subtraction the number 2010 is written, the maximum number of 7 that can be used, so that the total numbers used is a minimum is

(A) 284 (B) 286 (C) 288 (D) 290

3. In the adjoining rangoli design of the each four sided figures is rhombus and the distance between any two dots is 1 unit. The total area of the design is



(A) $36\sqrt{3}$ (B) $9\sqrt{3}$ (C) $24\sqrt{3}$ (D) $18\sqrt{3}$

4. In the addition problem shown, different letters represent different digits. If the carry over from adding units digit is 2, then (A+I) cannot be

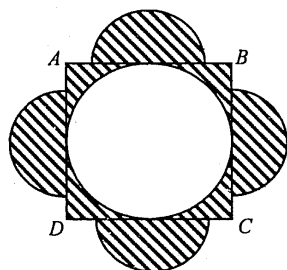
(A) 2 (B) 4
(C) 7 (D) 9

$$\begin{array}{r} \text{D I D} \\ \text{I} \\ \text{S A Y} \\ \hline \text{M A T H} \end{array}$$

5. The percentage of the natural numbers from 10 to 99 both inclusive which are the product of consecutive natural numbers is

(A) $9\frac{7}{9}$ (B) $7\frac{7}{9}$ (C) 10 (D) 9

6.



In the adjoining figure, $ABCD$ is a square of side 4 units. Semicircles are drawn outside the squares with diameter 2 units as shown. The area of the shaded portion in square units is

(A) 8 (B) 16 (C) $16-2\pi$ (D) $8-\pi$

7. $n = 1 + 11 + 111 + \dots + 1111111111$. The digit sum of n is

(A) 39 (B) 38 (C) 37 (D) 36

8. $A + B = C$, $B + C = D$, $D + A = E$ then $A + B + C$ is

(A) E (B) $D+E$ (C) $E-D$ (D) $B-D+C$

9. A three digit number $ab7 = a^3 + b^3 + 7^3$. Then a is

(A) 6 (B) 4 (C) 7 (D) 8

10. Among the participants of the screening test, some of them together got the correct answers for all the problems, not all of them got more than 8 problems correct. The maximum and minimum number of problems solved by the three together are

(A) 25, 25 (B) 24, 26 (C) 25, 27 (D) 25, 28

11. $\frac{1}{a} + \frac{1}{b} = \frac{1}{13}$ where a, b are natural numbers.

And (1) $a = b = 26$ (2) $a = 13, b = 13 \times 14$ (3) $a = 14, b = 13 \times 14$.

Of these statements the correct statements are

(A) (1) and (2) (B) (1) and (3) (C) (2) and (3) (D) (1)(2) and (3)

12. $a^3 + b^3 = p_1 \times p_2, a^3 - b^3 = p_3$ p_1, p_2, p_3 are prime numbers, $a > b, 2 \leq a, b \leq 5$

The following statements are also given:

- (1) $p_1 + p_2 + p_3$ is a prime number.
 (2) $p_3 - (p_1 + p_2)$ is a prime number.
 (3) $p_1 \sim p_2$ is a prime number.

Then the values of a & b are respectively.

- (A) (3,4) (B) (4,3) (C) (3,2) (D) (2,3)

13. p is a prime number greater than 3. when p^2 is divided by 12 the remainder is

- (A) always an odd number greater than 2.
 (B) always 1.
 (C) 1 or 11
 (D) always an even number.

14. Two regular polygons have the number of sides in the ratio 3:2 and the interior angles in the ratio 10:9 in that order. The number of sides of the polygons are respectively

- (A) 6 and 4 (B) 9 and 6 (C) 12 and 8 (D) 15 and 10

15. a is a real number such that $a^3 + 4a - 8 = 0$. Then the value of $a^7 + 64a^2$ is

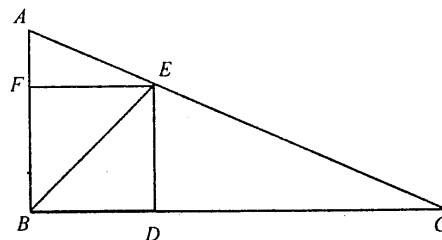
- (A) 128 (B) 164 (C) 256 (D) 180

PART – B

Note:

- Write the correct answer in the space provided in the response sheet.
- For each correct response you get 1 mark; for each incorrect response you lose $\frac{1}{2}$ mark.

16. ABC is a right angled triangle with $B=90^\circ$. $BDEF$ is a square. BE is perpendicular to AC . The measure of $\angle DEC$ is _____.



17. p is prime number and $p = a^2 - 1$. The number of divisors of $a+p$ is _____.

18. $a = 1+3+5+7+\dots+2009$
 $b = 2+4+6+8+\dots+2010$

then the value of $(a-b)^2$ is _____.

19. $1\frac{1}{2} + 2\frac{2}{3} + 3\frac{3}{4} + \dots + n\frac{n}{n+1}$ is on complete simplification has the denominator _____.

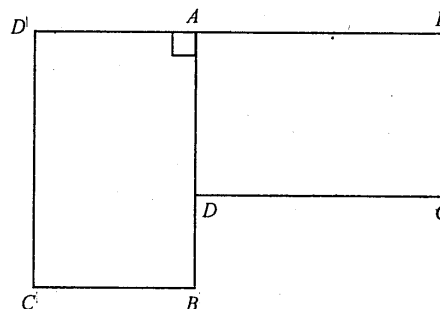
20. The biggest value of $\frac{10a}{10+a}$ ($a \in \mathbb{N}$) is never greater than _____.

21. From a point within an equilateral triangle perpendiculars are drawn to the three sides and are 5, 7 and 9 cms in length. The perimeter of the triangle is _____ cm.

22. The number of terms in the expansion $(a+b+c)^3$ is _____.

23. The value of x satisfying the equation $\frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{x}}}} = \frac{3}{4}$ is _____.

24. $ABCD$ is a rectangle rotated clockwise about A by 90° as shown. The rotation takes B to B' , C to C' , D to D' . $AB = 6\text{cm}$, $BC = 10\text{cm}$. The breadth of the rectangle $ABCD$ is _____.



25. AB is a line segment 2000cm long. The above design of semicircles are drawn on AB , with $AP = 5\text{cm}$ and repeating the designs. The area enclosed by the semicircular designs from A to B is _____.

